

JUAN SEBASTIÁN GÓMEZ Q.

Electronics Engineer & Electrical Engineering Ph.D. Member IEEE

@ juan.gomez@unab.cl

+56 9 4408 5906

https://jgomez-unab.github.io

orcid.org/0000-0002-6507-1462

SUMMARY

Juan Sebastián Gómez is an Associate Professor at the Faculty of Engineering and a researcher at the Energy Transformation Center, Universidad Andrés Bello, Santiago, Chile. He received the B.E. degree in Electronics Engineering from Universidad Distrital Francisco José de Caldas, Bogotá, Colombia, in 2011, and the Ph.D. degree in Electrical Engineering from the University of Chile in 2020. His research and teaching activities are primarily focused on automatic control, with emphasis on model-based predictive control, distributed and networked control, and cyber-physical systems. These methodologies are mainly applied to electrical engineering problems, particularly renewable energy integration in microgrids and smart grids, ancillary services, and power quality. He currently leads ANID-funded research as Principal Investigator of a Fondecyt Iniciación project and Director of a Fondecyt project on distributed cyber-physical emulation of smart electrical grids. He also serves as an Adjunct Researcher in an ANID-Anillo project on resilient agrivoltaic grids and at the Solar Energy Research Center (SERC Chile).

PUBLICATIONS

In Progress Publications

- 7 N. Yasic, I. Rivera, C. Polanco, F. Jaramillo-Montoya, and J. S. Gómez. "Fault location and classification in electrical grids through decision-tree-based algorithms". Submitted to IEEE-PES International Meeting 2027.
- 6 C. Burgos, D. Muñoz, A. Navas, and J. S. Gómez. "Distributed Model Predictive Control: An Overview on Tertiary and Secondary Control Layers for Microgrids". Submitted to IEEE Transactions on Industrial Informatics. 2026.
- 5 Enrique Espina, Claudio Burgos, J. S. Gómez, and Alex Navas. "Control and Operation of Single-/Three-Phase Four-Wire Microgrids: An Overview". Work in progress. 2026.
- 4 F. Manière, J. S. Gómez, and C. Parrado. "A Multi-Criteria Study Case of Statistical, Machine-Learning and Foundation Models for Building Energy Consumption Forecasting". Work in Progress. 2026.
- 3 C. Parrado and J. S. Gómez. "Quantifying the Dispatch Fidelity Gap in Utility-Scale PV+BESS: Receding-Horizon LP, Robust MPC, and Soft Actor-Critic under Atacama Market Conditions". Submitted to Advanced Energy and Sustainability Research. 2026.
- 2 D. Sosa, B. Castro, C. Carrasco, and J. S. Gómez. "Integration of Energy Management Systems (EMS) for Agrivoltaic Farming and Desalination Applications, Review". Work in Progress. 2026.
- 1 Daniel Sosa, Luis G. Marín, José Luis Uribe, Alex Navas, Oscar Cartagena, J. S. Gómez, José Vuelvas, and Ana Llor. "Transactive Microgrid with Distributed Energy Resources Based on Blockchain Technology". Submitted in Sustainable Energy, Grids and Networks. 2025.

Journal Articles

- 117 M. Bravo-López, A. Garces, W. Gil-González, J. S. Gómez, C. González-Castaño, and C. Restrepo. "Nonlinear Model Predictive Wide-Area Frequency Control for Inverter-Dominated Systems with IEC 61850-Based Communication Validation". In: *IEEE Access* (2026). DOI: 10.1109/ACCESS.2026.3695782.
- 116 C. Muñoz-Olivar, J. S. Gómez, F. Pastrana Rendón, C. Taramasco-Toro, M. Sánchez-Cárdenas, and C. J. Avendaño-Vásquez. "Development and validation of the synthetic index of palliative care integration into primary health care". In: *BMC Health Services Research* (2026). DOI: 10.1186/s12913-026-15191-w.
- 115 C. Muñoz-Olivar, J. S. Gómez, F. Pastrana Rendón, C. Taramasco-Toro, M. Sánchez-Cárdenas, and C. J. Avendaño-Vásquez. "Indicators for palliative care integration in primary health care: how expert profile shapes consensus in a modified Delphi study". In: *BMC Health Services Research* (2026). DOI: 10.1186/s12913-026-14781-y.
- 114 C. Muñoz-Olivar, F. Marquez-Doren, J. S. Gómez, C. Taramasco-Toro, and C. J. Avendaño-Vásquez. "Expert Perspectives on Integrating Palliative Care into Primary Health Care: A Qualitative Analysis of a Modified Delphi Study". In: *Nursing Reports* 16.1 (2026). DOI: 10.3390/nursrep16010020. URL: https://www.mdpi.com/2039-4403/16/1/20.

IN PROGRESS TOPICS

- Ancillary Services Between Microgrids and Power Grids
- Distributed Control Applications in Microgrids
- Cyberphysical Systems applications in Industry Electrification
- Stability Analysis of Microgrids with Distributed Predictive Control Schemes.

EDUCATION

Ph.D. in Electrical Engineering

2014 – 2020

University of Chile

B.E in Electronics Engineering

2004 – 2011

Universidad Distrital FJdC

PROFESSIONAL COMMITMENTS

- Signatory, Leiden Declaration on Artificial Intelligence and Mathematics (2026)

LANGUAGES

Spanish (Native)

English

INTERNSHIPS

University of Nottingham - Visiting Researcher

Aug 2023

- Invited by the Power Electronics, Machines and Control Group (PEMC).
- University of Nottingham, University Park, Nottingham, NG7 2RD.
- Work in progress with Prof. Mark Sumner and Prof. Marco Rivera.
- Funds from Research and Doctorate Vice Rectory Projects- Universidad Andrés Bello.

University of Waterloo - International Visiting Graduate Student

Oct 2017 – Feb 2018

- Internship Supervisor: Prof. John Simpson-Porco.
- University of Waterloo, 200 University Avenue West. Waterloo, Ontario - Canada.
- Internship supported by CONICYT-PCHA/Doctorado Nacional para extranjeros/2015- 21150555, FONDECYT 1170683 grant and, Complex Engineering Systems Institute (CONICYT – PIA – FB0816).

LIST OF PUBLICATIONS



Google Scholar

- ^{j13} B. Oliveira, J. S. Gómez, J. C. Cebrian, L. A. de Oro Arenas, and H. K. Morales. "Technical Aspects for Smart Grids Interoperability: IoT and IEC 61850 as Key Enablers". In: *IEEE ACCESS* (2026), pp. 52440–52457. DOI: 10.1109/ACCESS.2026.3680294.
- ^{j12} J. S. Gómez, L. G. Marín, H. K. Morales, A. Llor, V. Hincapié, J. Vuelvas, A. Navas, C. Burgos, L. S. de Lima Tibúrcio, P. F. Silva, L. A. de Oro Arenas, A. Azar I. Rivera, J. Valencia, N. Yasic, F. Flores-Bahamonde, G. Aguila, and J. C. Cebrian Amasifen. "Fuelling the Transition: A Comprehensive Analysis of Hydrogen Roadmaps in Chile, Brazil and Colombia". In: *Renewable and Sustainable Energy Reviews* (2025). DOI: 10.1016/j.rser.2025.115885.
- ^{j11} C. Muñoz-Olivar, M. Pineiro, J. S. Gómez, C. J. Avendaño-Vásquez, P. Ormeño-Arriagada, S. Palma-Rivadeneira, and C. Taramasco-Toro. "Education and Symptom Reporting in an mHealth App for Patients With Cancer: Mixed Methods Development and Validation Study". In: *JMIR Human Factors* 12 (2025), pp. 1–13. DOI: 10.2196/60169.
- ^{j10} R. Pérez, M. Rivera, B. Araya, J. S. Gómez, Y. Salgueiro, C. Restrepo, P. Wheeler, M. You, and M. Sumner. "Enhanced Microgrid Reliability through Software-Defined Networking and Extended Horizon Predictive Control". In: *Sustainable Energy, Grids and Networks* 42 (2025). DOI: 10.1016/j.segan.2025.101635.
- ^{j9} A. Navas, C. Burgos, J. S. Gómez, E. Espina, J. Llanos, D. Sáez, M. Sumner, and D. Olivares. "Distributed predictive secondary control with soft constraints for optimal dispatch in hybrid AC/DC microgrids". In: *IEEE Transactions on Smart Grid* 14.6 (2023), pp. 4204–4218. DOI: 10.1109/TSG.2023.3261569.
- ^{j8} J. S. Gómez, A. Navas, F. Flores, L. Tarisciotti, C. Garcia, F. Nuñez, J. Rodriguez, and A. Z. Cipriano. "Predictive Control for Current Distortion Mitigation in Mining Power Grids". In: *Applied Sciences MPDI - Special Issue Multiport Converters for Renewable Energy Sources and EV Applications* 13.3523 (2023). DOI: 10.3390/app13063523.
- ^{j7} M. Martínez-Gómez, C. Burgos, H. K. Morales-Paredes, J. S. Gómez, A. K. Verma, and J. P. Bonaldo. "Distributed Cooperative Control of Clusters of Power Quality Compensators based on the Conservative Power Theory for Grid-tied AC Microgrids". In: *Sustainability* 15.22 (2023), p. 15698. DOI: 10.3390/su152215698.
- ^{j6} E. Rute, A. Navas, J. S. Gómez, E. Espina, C. Burgos, D. Sáez, M. Sumner, and D. Muñoz. "Distributed Predictive Control for Secondary Level in Hybrid ac/dc Microgrids". In: *IEEE Journal of Emerging and Selected Topics in Power Electronics* (2022). DOI: 10.1109/JESTPE.2022.3157979.
- ^{j5} A. Navas, C. Burgos, J. S. Gómez, F. Donoso, L. Tarisciotti, D. Sáez, R. Cárdenas, and M. Sumner. "Distributed predictive secondary control for imbalance sharing in AC microgrids". In: *IEEE Transactions on Smart Grid* (2021). DOI: 10.1109/TSG.2021.3108677.
- ^{j4} A. Navas, J. S. Gómez, J. Llanos, E. Rute, D. Sáez, and M. Sumner. "Distributed Predictive Control Strategy for Frequency Restoration of Microgrids Considering Optimal Dispatch". In: *IEEE Transactions on Smart Grid* (2021). DOI: 10.1109/TSG.2021.3053092.
- ^{j3} I. Serban, S. Céspedes, C. Marinescu, C. Azurdia-Meza, J. S. Gómez, and D. Sáez. "Communication Requirements in Microgrids: a Practical Survey". In: *IEEE Access* (Mar. 2020). DOI: 10.1109/ACCESS.2020.2977928.
- ^{j2} J. S. Gómez, J. Rodriguez, C. Garcia, L. Tarisciotti, F. Flores, J. Pereda, F. Nuñez, A. Z. Cipriano, and J. C. Salas. "An Overview of Microgrids Challenges in the Mining Industry". In: *IEEE Access* (2020). DOI: 10.1109/ACCESS.2020.3032281.
- ^{j1} J. S. Gómez, D. Sáez, J. W. Simpson-Porco, and R. Cárdenas. "Distributed Predictive Control for Frequency and Voltage Regulation in Microgrids". In: *IEEE Transactions on Smart Grid* 11.2 (Mar. 2020), pp. 1319–1329. DOI: 10.1109/TSG.2019.2935977.



Conferences

- ^{c12} Diego Montoya, Natalia Jara, J. S. Gómez, Carlos Restrepo, and Catalina González-Castaño. "A GIS-Based Method to Prioritize MV Feeders for PV-SDGF Interconnection". In: *17th IEEE International Symposium on Power Electronics for Distributed Generation Systems (PEDG)*. In Press. 2026.
- ^{c11} J. S. Gómez, A. Azar, A. Ulloa, N. Yasic, C. Carrasco, and C. Parrado. "Dynamic Hydrogen Microgrid Control: A Dual-Domain MPC Approach with Local Flow Loops". In: *2025 IEEE-CHILEAN Conference on Electrical, Electronics Eng., Information and Communication Technologies (CHILECON)*. 2025. DOI: 10.1109/CHILECON66915.2025.11476106.
- ^{c10} N. Yasic, A. Ulloa, C. Carrasco, C. Parrado, and J. S. Gómez. "Hardware-in-the-Loop Validation of a Basic PV Model Using OpenModelica and OPAL-RT". In: *2025 IEEE CHILEAN Conference on Electrical, Electronics Engineering, Information and Communication Technologies (CHILECON)*. 2025. DOI: 10.1109/CHILECON66915.2025.11476587.

PROJECTS, GRANTS & SCHOLARSHIPS

- Adjunct Researcher - ANID/CIN250043: "Solar Energy Research Center" (SERC-Chile). RL5- Smart Integrated Solar Systems. ANID-National Interest Center (2026-2030). **Work in Progress**
- Director - Research project EQM250011: "Platform for distributed cyber-physical emulation of smart electrical grids". ANID-Fondequip project (2026-2028). **Work in Progress**
- Adjunct Researcher - Research project ATE250063: "High efficiency and resilient agriphotovoltaic grid for the energization of water desalination systems". ANID-Anillo project (2026-2028). Director: Freddy Flores-Bahamonde. **Work in Progress**
- Researcher - Research project DI-08-24/IA: "Energía Inteligente: Gemelos Digitales y IA para la Predicción de Fallos en Redes Eléctricas". UNAB Endowment- Concurso inteligencia artificial para la I+D 2024 (2025). Principal Researcher: Cristobal Parrado **Closed**
- Researcher and International Project Coordinator - Research project: "Energy Management System for Hydrogen-based Flexibility Services". Climat AmSud (France) (2024-2025). **Closed**
- Principal Researcher - Research project: "Ancillary Services Integration in Microgrids for Smart Grid Supporting". Fondecyt - Iniciación (Chile) (2024-2026). **Work in Progress**
- Associate Researcher - Research project: "Control and Optimization of Multiport DC-DC Converters for Integration of Short-scale Distributed Generation and Electrical Vehicles in Residential Areas". Escuela Superior Politécnica de Chimborazo - Universidad de las Fuerzas Armadas-ESPE (Ecuador) 2024. Principal Researcher: Dr. Jorge Ambato. **Closed**
- Principal Researcher - Research project: "Ancillary Services Management for Microgrids Integration to the Smart Grid". Universidad Andrés Bello (Chile) (2023-2024). Associate Researchers: Dr. Seyed Alireza Davari, Dr. Freddy Flores-Bahamonde. **Closed**
- Associate Researcher - Research project: "Optimal Energy Management Systems for Hybrid Generation Systems". Universidad de las Fuerzas Armadas- ESPE (Ecuador) (2023). Principal Researcher: Dra. Jacqueline Llanos. **Closed**
- Postdoctoral position - Anillo project ACT192013: "Cyber-physical systems for smart mining: Industrial electronics 4.0 and data-driven process control" ANID PIA ACT192013. **Closed**
- National Doctoral Scholarship for foreigners - National Council for Scientific and Technological Research: CONICYT-PCHA/Doctorado Nacional para extranjeros/2015- 21150555. **Closed**

- c9 K. Chamorro-Caceres, A. Navas-Fonseca, J. S. Gómez, C. Ahumada, O. Cartagena, L. Marin, Doris Saez, and G. Aguila. "Hierarchical energy management system with demand side management for green hydrogen-based microgrids". In: *IEEE 42 Central America and Panama Convention (CONCAPAN)*. 2024, pp. 1–6. DOI: 10.1109/CONCAPAN63470. 2024. 10933606.
- c8 Sofia Intriago, Paola Robayo, Jacqueline Llanos, Franklin Silva, and J. S. Gómez. "Comparison of Control Strategies for Monitoring the Maximum Power Point Tracking of a Photovoltaic Plant". In: *2022 IEEE Sixth Ecuador Technical Chapters Meeting (ETCM)*. 2022, pp. 1–6. DOI: 10.1109/ETCM56276. 2022. 9935710.
- c7 A. Navas, C. Burgos, E. Espina, E. Rute, J. S. Gómez, D. Sáez, and M. Sumner. "Distributed Predictive Secondary Control for Voltage Restoration and Economic Dispatch for DC Microgrids". In: *2021 4th IEEE International Conference on DC Microgrids (ICDCM 2021)*. Arlington, VA, USA: IEEE, July 2021. DOI: 10.1109/ICDCM50975. 2021. 9504612.
- c6 A. Navas, C. Burgos, J. S. Gómez, J. Llanos, E. Espina, D. Sáez, and M. Sumner. "Distributed Predictive Control using Frequency and Voltage Soft Constraints in AC Microgrids including Economic Dispatch of Generation". In: *IECON 2021 47th Annual Conference of the IEEE Industrial Electronics Society*. 2021, pp. 1–7. DOI: 10.1109/IECON48115. 2021. 9589500.
- c5 E. Espina, A. Navas, J. S. Gómez, D. Sáez, and R. Cárdenas. "Experimental Performance Evaluation of a Distributed Secondary Control Strategy for Hybrid ac/dc-Microgrids in the event of Communication Loss/Delay". In: *2021 23rd European Conference on Power Electronics and Applications (EPE '21 ECCE Europe)*. Virtual Conference: IEEE, Sept. 2021. DOI: 10.23919/EPE21ECCEEurope50061. 2021. 9570572.
- c4 H. K. Morales-Paredes., C. Burgos, J. P. Bonaldo, D. T. Rodrigues, and J. S. Gómez. "Cooperative Control of Power Quality Compensators in Microgrids". In: *2021 IEEE Green Technologies Conference (GreenTech)*. 2021, pp. 380–386. DOI: 10.1109/GreenTech48523. 2021. 00066.
- c3 J. S. Gómez, J. Llanos, C. Burgos, E. Espina, and J. Rodriguez. "Cooperative Power Conditioners for Microgrids in Mining". In: *2021 23rd European Conference on Power Electronics and Applications (EPE '21 ECCE Europe)*. Virtual Conference: IEEE, Sept. 2021. DOI: 10.23919/EPE21ECCEEurope50061. 2021. 9570651.
- c2 E. Espina, C. Burgos, J. S. Gómez, J. Llanos, E. Rute, A. Navas, M. Martinez, D. Sáez, and R. Cárdenas. "Experimental Hybrid ac/dc-Microgrid Prototype for Laboratory Research". In: *2020 22nd European Conference on Power Electronics and Applications (EPE '20 ECCE Europe)*. IEEE, Sept. 2020, P.1–P.9. DOI: 10.23919/EPE20ECCEEurope43536. 2020. 9215751.
- c1 J. Llanos, J. S. Gómez, D. Sáez, D. Olivares, and J. W. Simpson-Porco. "Economic Dispatch by Secondary Distributed Control in Microgrids". In: *2019 21st European Conference on Power Electronics and Applications (EPE '19 ECCE Europe)*. Genova, Italy: IEEE, Sept. 2019, P.1–P.10. DOI: 10.23919/EPE. 2019. 8915499.

Thesis

- t1 J. S. Gómez. "Distributed predictive control for frequency and voltage regulation in microgrids". University of Chile - Electrical Engineering Department, 2020. URL: <http://repositorio.uchile.cl/handle/2250/173842>.

EXPERIENCE

Researcher Professor - Universidad Andrés Bello

 July 2022 - On going

- Researcher at the Energy Transformation Center - Universidad Andrés Bello.
- Teaching post and undergraduate students on topics such control and power systems modelling and smart grids.

Lecturer - Seminar on Power Systems

 First term 2022

- Teaching post and undergraduate students on topics such as synchronous machine theory and, power system stability (long term, short term and transient stability), power systems modelling and smart grids.
- Pontificia Universidad Católica de Chile.

COMMITTEE MEMBER

- "Comparación de Eficiencia de Conversión de Energía del Procesamiento Total y Parcial de Potencia Basado en el Convertidor Buck-Boost Versátil" MSc. Thesis on Engineering Sciences- Energy Conversion - Esteban Ibáñez Muñoz. Universidad de Talca. Supervisor: C. Restrepo. 2026.
- "Control Distribuido No-Cooperativo de Convertidores de Potencia para Micro-redes Híbridas AC/DC" M.Sc. Thesis on Engineering Sciences- Tomás de la Fuente Molina. Universidad de Santiago de Chile (USACH). Supervisor: E. Espina. 2024.
- "Comparación de Eficiencia de Conversión de Energía del Procesamiento Total y Parcial de Potencia Basado en el Convertidor Buck-Boost Versátil" Eng. Thesis on Electronics- Esteban Ibáñez Muñoz. Universidad de Talca. Supervisor: C. Restrepo. 2024.
- "Enhancing Hierarchical Microgrid Resilience with FCS-MPC and Software-Defined Networking" Ph.D. Thesis on Systems of Engineering- Ricardo Pérez Guzmán. Universidad de Talca. Supervisor: M. Rivera. 2024.
- "Desempeño de la interoperabilidad, escalabilidad y seguridad en las comunicaciones de una microrred de energía eléctrica" M.Sc. Thesis on Industrial Automation- Jair Hernán Vargas Díaz. Universidad Nacional de Colombia. Supervisor: H. D. Zapata. 2022.

SUPERVISION

- "Método basado en Sistemas de Información Geográfica para la priorización de alimentadores de media tensión para la interconexión de PMGD fotovoltaicos"-Co-Guide. Graduation Work- Civil Electrical Engineering - N. Jara. Universidad de Talca. 2026.
- "Control Predictivo Distribuido Basado en Modelo de Voltaje y Potencia de Redes Eléctricas DC"-Guide. Graduation Work- Robotics and Automation Engineering - N. Miranda. Universidad Andrés Bello. 2026.
- "Unidad Medidora de Fasores Acelerada por Cuda"-Guide. Graduation Work- Robotics and Automation Engineering - A. Cartillo. Universidad Andrés Bello. 2026.
- "Diagnóstico de Fallas en Redes Eléctricas"-Guide. Graduation Work- Robotics and Automation Engineering - I. Rivera. Universidad Andrés Bello. 2026.
- "Gemelo Digital de Planta Fotovoltaica Estructurado por Estandar IEC 61850"-Guide. Graduation Work- Robotics and Automation Engineering - N. Yasic. Universidad Andrés Bello. 2025.
- "Optimización de Operación de OLTC para Prolongar su Vida Útil con Control Predictivo"-Guide. Graduation Work- Robotics and Automation Engineering - A. Ulloa. Universidad Andrés Bello. 2025.
- "Sistema de Gestión de Energía y Oxígeno Hospitalario"-Guide. Graduation Work- Robotics and Automation Engineering - A. Azar. Universidad Andrés Bello. 2025.
- "Técnica de localización óptima de electrolíneas basada en problemas de ruteo de vehículos eléctricos"- Co-Guide. Graduation Work- Electrical Engineering - G. Fuentes. Universidad de O'Higgins. Supervisor: C. Burgos. Chile 2023.
- "Planificación basada en datos de puntos de carga para transporte terrestre"- Guide. Graduation Work- Magister on Information and Telecommunication Technologies Management - C. Correa. Universidad Andrés Bello. Co-Guide: F. Flores-Bahamonde. Chile 2023.
- "Diseño y simulación de una estrategia de control avanzada para el seguimiento de punto de máxima potencia de una planta fotovoltaica"- External Collaborator. Graduation Thesis of Engineering degree in electronics and instrumentation - S. Intriago and P. Robayo. Universidad de las Fuerzas Armadas, ESPE. Supervisor: J. Llanos. Ecuador 2022.

Postdoctoral Researcher - Pontificia Universidad Católica de Chile

📅 April 2020 – December 2022

- Research project: Anillo project ACT192013: "Cyber-physical systems for smart mining: Industrial electronics 4.0 and data-driven process control".
 - Development of control strategies for microgrids applications in mining industries.
 - Cooperation project: Pontificia Universidad Católica de Chile - Universidad Andrés Bello.
 - This work was partially funded by ANID under grant ANID PIA ACT192013.
-

External Consultant - Rodrigo Aristizabal Cia. S.A. - RAYCO S.A.

📅 Jun 2016 – 2021

- Consultant for the arising corporate Research, Innovation, and Development team.
 - SCADA instructor for junior engineers.
 - External consultant for engineering design, and engineering supervision for automation and control projects related to oil and gas industries, and water treatment plants.
-

Ph.D. Student - University of Chile

📅 Mar 2014 – Jan 2020

- Research project: "Distributed Predictive Control for Frequency and Voltage Regulation in Micro-grids".
 - Ph.D. student affiliated with the Microgrids Control Lab. at the University of Chile in the Department of Electrical Engineering.
 - Development of secondary control strategies for voltage and frequency regulation in microgrids, preserving active and reactive power consensus, based on distributed predictive control considering communication issues among distributed generators.
 - Simulator development and experimental setup.
 - This research has been supported by a Ph.D. scholarship from the CONICYT-PCHA/Doctorado Nacional para extranjeros/2015- 21150555, the FONDECYT 1170683 grant, Complex Engineering Systems Institute (CONICYT – PIA – FB0816), Solar Energy Research Center SERC-Chile (CONICYT/FONDAP/ Project under Grant 15110019) and the NSERC Discovery Grant RGPIN- 2017-04008.
 - Advisor: Prof. Doris Sáez Dr. Eng.
 - Reviewers: Prof. Josep Guerrero Ph.D., Prof. Daniel Olivares Ph.D., Prof. Cesar Azurdia Ph.D.
-

Recitation - Dynamical Phenomena in Electrical Grids

📅 First term 2015

- Teaching assistance and support of under/postgraduate students in topics such as synchronous machine theory and, power system stability (long term, short term and transient stability).
 - Lecturer: Prof. Claudia Rahmann Ph.D.
-

Specialist Automation Engineer - Rodrigo Aristizabal Cia. S.A. - RAYCO S.A.

📅 Jan 2013 – Mar 2016

- Management and coordination of multidisciplinary teams for development of technical solutions for oil and gas industries (EPC projects).
 - Decision support on process operation and maintenance, project scope, networking, installation, start-up and maintenance schedules.
 - Projects developed to Gases de Occidente, ECOPETROL S.A. - Tanker Load Units- Coveñas Maritime port, HOCOL S.A. - Ocelote-Palmeras pipeline, among others.
-

Project Engineer - Rodrigo Aristizabal Cia. S.A. - RAYCO S.A.

📅 Jun 2010 – Jan 2013

- Design, development, maintenance and documentation of automation and communication projects for Energy and, oil and gas industries.
 - Projects developed to Petrobras Colombia - Guando oil field, EMGESA, HOCOL, S.A., ECOPETROL S.A. (Tras-andinean oil pipeline, Coveñas Maritime port, Pozos colorados Maritime port), Oleoducto Central S.A. (OCENSA), among others.
-

TEACHING

- Control II (Lecturer) - Undergraduate; Universidad Andrés Bello 2026-On going.
- Advanced Control (Lecturer) - Undergraduate; Universidad Andrés Bello 2023-On going.
- Control Theory / Control I (Lecturer) - Undergraduate; Universidad Andrés Bello 2023-On going.
- Transformation and Control of Electric Energy (Lecturer) - Doctorate; Universidad Andrés Bello 2023-2024.
- Seminar on Power Electrical Systems (Lecturer) - Post/Undergraduate; Pontificia Universidad Católica de Chile 2022.
- SCADA and automation engineering training (Instructor)- Junior engineers; Rodrigo Aristizabal S.A. 2016-On going
- Dynamical Phenomena in Electrical Grids (Recitation) - Post/Undergraduate; University of Chile 2015.

PEER REVIEWER FOR

- IEEE Transactions on Smart Grid
- CHILECON
- IEEE Access
- IEEE Transactions on Sustainable Energy
- Revista Facultad de Ingeniería - U de Antioquia
- Fondecyt Research Init. Proj. Competition ANID - Chile